

Date .

Unit-1

Q. Let (x, y) be discrete bivariate r.v with joint PMF
 $P(x=x, y=y) = k(2x+y)$
 $x=1, 2 \quad y=1, 2$

Marginal of x

$$P(X=1) = 7/18$$

$$P(X=2) = 11/18$$

x	1	2
$P(X=x)$	7/18	11/18

1) k 2) obtain marginal of x & y 3) $E(X)$, $E(Y)$, $E(XY)$ 4) $SD(X)$, $SD(Y)$ 5) Find $P(X=x | Y=2)$

6) Check independence.

Marginal of y

$$P(Y=1) = 8/18$$

$$P(Y=2) = 10/18$$

y	1	2
$P(Y=y)$	8/18	10/18

Ans: $\sum_{x=1}^2 \sum_{y=1}^2 P(X=x, Y=y) = 1$

3) $E(X) = \sum_{x=1}^2 x P(X=x)$

sub $x=1 \quad y=2$
 $3k + 4k + 5k + 6k$
 $= 18k = 1$
 $k = 1/18$

$$E(X) = 1 \times \frac{7}{18} + 2 \times \frac{11}{18} = \frac{29}{18}$$

$$E(Y) = 1 \times \frac{8}{18} + 2 \times \frac{10}{18} = \frac{28}{18}$$

Marginals of x & y

2)

x/y	1	2
1	3/18	4/18
2	5/18	6/18

$$= 7$$

$$= 11$$

8

10

$$E(XY) = \sum_{x=1}^2 \sum_{y=1}^2 xy P(X=x, Y=y)$$

$$1 \times 1 \times \frac{3}{18} + 1 \times 2 \times \frac{4}{18} +$$

$$2 \times 1 \times \frac{5}{18} + 2 \times 2 \times \frac{6}{18} = \frac{45}{18}$$

$$4) SD(X) = \sqrt{V(X)}$$

$$V(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \sum x^2 p(x)$$

$$\frac{4}{18} + \frac{6}{18} = 1 //$$

$$\frac{10}{18} \quad \frac{10}{18}$$

$$E(X^2) = 1^2 \times \frac{1}{18} + 2^2 \times \frac{11}{18}$$

6) check Independence.

Correlation.

$$Cov(X, Y) = \frac{Cov(X, Y)}{\sigma_X \cdot \sigma_Y}$$

$$V(X) = \frac{51}{18} - \left[\frac{29}{18} \right]^2$$

$$= \frac{77}{324}$$

$$Cov(X, Y) = E(XY) - E(X) \cdot E(Y)$$

$$\frac{45}{18} - \frac{29}{18} \cdot \frac{28}{18}$$

$$= -0.006 //$$

$$\frac{\sqrt{77}}{324} = 0.487 //$$

No relation since zero

$$SD(Y) =$$

Continuous.

$$E(Y^2) = \sum y^2 p(y)$$

Q2) The fraction x of male runners and the fraction y of female runners who

$$1^2 \times \frac{8}{18} + 2^2 \times \frac{10}{18} = \frac{48}{18}$$

Complete marathon race is described by the JPDF

$$V(X) = \frac{48}{18} - \left(\frac{28}{18} \right)^2 = \frac{20}{81} //$$

$$f(x, y) = \begin{cases} kxy & 0 < x < 1 \\ & 0 < y < 1 \end{cases}$$

$$\frac{\sqrt{20}}{18} = 0.497 //$$

$$5) \frac{P(x=1 | y=2) + P(x=2 | y=2)}{P(y=2)}$$

- 1) k.
- 2) Marginals
- 3) $E(x)$, $E(y)$, $E(xy)$
- 4) $\text{Cov}(x, y)$
- 5) check Independence.

$$D \int_0^1 \int_0^x kxy \, dx \, dy$$

double $\int$

$$\int_0^1 kxy \left[\frac{y^2}{2} \right]_0^x dx$$

$$k \int_0^1 \frac{x^3}{2} dx = \left[\frac{kx^4}{4 \times 2} \right]_0^1$$

$$k \times \frac{1}{8} = k = 8$$

- 2) Marginals.

$$f(x) = \int_0^x 8xy \, dy$$

$$8x \left[\frac{y^2}{2} \right]_0^x = 4x^3, \quad 0 < x < 1$$

$$f(y) = \int_0^1 8xy \, dx$$

$$8 \frac{x^2}{2} y = 4x^2 y$$

$$\int_0^1 4x^2 y \cdot 4y(1-y^2) \, dy$$

$$E(x) = \int_0^1 x f(x) \, dx$$

$$\int_0^1 4x^4 \, dx = \int_0^1 4x^3 \times x \, dx = \int_0^1 4x^4 \, dx = \left[\frac{4x^5}{5} \right]_0^1 = \frac{4}{5}$$

$$E(y) = \int_0^1 y f(y) \, dy$$

$$= \int_0^1 4y(1-y)^2 \, dy$$

$$= \frac{8}{15}$$

$$E(xy) = \int_0^1 \int_0^x xy f(x, y) \, dy \, dx$$

$$\int_0^1 \int_0^x xy 8xy \, dy \, dx$$

$$= \int_0^1 \int_0^x 8x^2 \left[\frac{y^3}{3} \right]_0^x \, dx = \frac{8}{3} \left[\frac{x^6}{6} \right]_0^1 = \frac{4}{9}$$

$$5) \text{Cov}(x, y) = E(xy) - E(x) \cdot E(y)$$

$$= \frac{4}{9} - \frac{4}{5} \cdot \frac{8}{15}$$

$$\neq 0$$

Bayes theorem

$$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

$$P(B/D) = \frac{P(D/B) \cdot P(B)}{P(D)}$$

$$= \frac{0.05 \cdot 0.04}{0.038} = 0.5263$$

52%

$$P(D) = P(D/A) \cdot P(A) + P(D/B) \cdot P(B)$$

$$= 0.038 //$$

Qa. Factory A produces 60% of bulbs and FB. 40% of bulbs 3% of FB. Bulbs from A are

defected 5% from B are defected. A randomly selective bulbs & defected and find the probability that it came from

the F.B — $P(B/D)$

Defective

FA 60% 3%

FB 40% 5%

$$P(A) = 0.6$$

$$P(B) = 0.4$$

$$P(D/A) = 0.03$$

$$P(D/B) = 0.05$$

Q2) In a college 70% of students attend regularly 30% attend irregularly 90% of regular students passed 50% of irregular 90% of students passed exam and find the prob the student attended regularly.

$$P(R) = 0.7$$

$$P(I) = 0.3$$

$$P(P/R) = 0.9$$

$$P(P/I) = 0.5$$

$$P(R/P) = \frac{P(P/R) \cdot P(R)}{P(P)}$$

$$0.08 // 80%$$

10 marks duration.

$$P(CP) = P(CP/R) \cdot P(R) + P(CP/B) \cdot P(B)$$

$$P(CP/B) = 0.78011$$

A urn A consist of 4 Red and 6 blue ball urn B

7R and 3 B 1 urn is chosen at random

and red ball is drawn

find the probability that

the ball came from urn B.

u1

u2

urn 1

urn 2

4R

7R

6B

3B

$$P(R/u_1) = 4/10$$

$$P(B/u_1) = 6/10$$

$$P(R/u_2) = 7/10$$

$$P(B/u_2) = 3/10$$

$$P(B) = 0.5$$

$$P(A) = 0.5$$

we take random
Prob we don't
know which
came

$$P(R) = P(R/u_1) \cdot P(u_1) +$$

$$P(R/u_2) \cdot P(u_2)$$

$$= 0.55011$$

$$P(B/R) = \frac{P(R/B) \cdot P(B)}{P(R)}$$

$$= 0.6311$$

Conjugate priors:-

Bernoulli likelihood &

Beta (prior) distribution

$$\text{posterior} = \text{likelihood} \times \text{prior}$$

Bernoulli PMF

$$P(X=x) = p^x q^{n-x}$$

$$\theta^x (1-\theta)^{n-x} \quad \left(\theta \in (0,1) \right)$$

If we want independent observation

$P = \{x_1, x_2, \dots, x_n\}$ are

drawn from a Bernoulli

then the likelihood is given

$$P(D/\theta) = \theta^{x_1} (1-\theta)^{1-x_1} \times \theta^{x_2} (1-\theta)^{1-x_2}$$

$$\prod_{i=1}^n \theta^{x_i} (1-\theta)^{1-x_i}$$

$$= \theta^{\sum x_i} (1-\theta)^{n - \sum x_i}$$

$$\theta^{\sum x_i} (1-\theta)^{n - \sum x_i}$$

$$\theta^{\sum x_i} (1-\theta)^{n - \sum x_i}$$

$$P(D/\theta) = \theta^k (1-\theta)^{n-k}$$

likelihood of

Bernoulli

Beta:

$$f(\theta) = \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

checking if we get
Beta poster.

$P(\theta/D)$ = likel $\times$ prior

$$\theta^k (1-\theta)^{n-k} \times \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

$$P(\theta/D) = \theta^{k+\alpha-1} (1-\theta)^{n+\beta-k-1}$$

Same//

$$P(\theta/D) = \text{Beta}(\alpha+k, n+\beta-k)$$

The Beta distribution is Conjugate.
prior to Bernoulli.

Binomial likelihood Beta prior

prior poster

no. of success

$$n \binom{n}{k} p^k q^{n-k}$$

no. of trial

$$P(X=x) = n \binom{n}{k} p^k q^{n-k}$$

$$P(D/\theta) = n \binom{n}{k} p^k q^{n-k}$$

$$f(\theta) = \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

$$P(P/\theta) = n \binom{n}{k} p^{k+\alpha-1} (1-p)^{n+\beta-k-1}$$

It's same

Poisson (likelihood) & gamma prior

PMF

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad (x=0, 1, \dots, n)$$

let m obs.

$$D = \{x_1, x_2, \dots, x_n\}$$

$$L.F = \frac{e^{-\lambda} \lambda^{x_1}}{x_1!} \times \frac{e^{-\lambda} \lambda^{x_2}}{x_2!}$$

$$P(D/\lambda) = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\sum x_i!}$$

$$P(D/\lambda) = \alpha e^{-n\lambda} \lambda^{\sum x_i}$$

$\hookrightarrow$ likelihood of pairs

$$P(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$$

$\lambda \sim q(\lambda, \beta)$

poisson mean, rename λ

$$E(X) = \frac{\alpha}{\beta}, \quad V(X) = \frac{\alpha}{\beta^2}$$

$$P(\lambda) = \alpha \lambda^{\alpha-1} e^{-\beta\lambda}$$

$$P(\lambda/D) = P(D/\lambda) \times P(\lambda)$$

$$e^{-n\lambda} \lambda^{n+1} \times \lambda^{\alpha-1} e^{-\beta\lambda}$$

$$= e^{-\lambda(n+\beta)} \lambda^{n+\alpha-1}$$

$$\lambda \sim q(\alpha + \epsilon n, n + \beta)$$

Exponential family

A distribution belongs to exponential family if its probability fn. can be written in the form

$$f(x, \theta) = h(x) e^{\eta(\theta) T(x) - A(\theta)}$$

$\eta(\theta)$ is natural parameter

where $h(x)$ depends only on data

$T(x)$ is sufficient statistic

$\eta(\theta)$ = natural parameter

A(0) normalizing fn.

Q) Bernoulli check belong to expo family or not

$$PMF = p^n q^{1-n} (n, 0, 1)$$

$$(1-p) \left(\frac{p}{1-p} \right)^n$$

$$\exp \left[\log(1-p) + n \log \frac{p}{1-p} \right]$$

$$\exp \left[(\log(1-p) + n \log \frac{p}{1-p}) \right]$$

$$1 \times \exp \left[n \log \left(\frac{p}{1-p} \right) + \log(1-p) \right]$$

Belongs to expo family

Q2) Binomial checking

$$P(X=x) = \binom{n}{x} p^x q^{n-x}$$

$$\binom{n}{x} \exp \left[x \log p + (n-x) \log(1-p) \right]$$

$$h(x) = \binom{n}{x} \exp \left[x \log \frac{p}{1-p} \right] + n \log(1-p)$$

5) Poisson check

$$P(n=1) = \frac{e^{-\lambda} \lambda^n}{n!}$$

$$f(n, \theta) = h(n) \times \exp$$

$$\frac{1}{n!} \exp[n \log \lambda - \lambda]$$

$$h(n) = \frac{1}{n!}$$

$$n \log = T(n)$$

$$q(n) = \lambda$$

$$-\lambda = \eta(\theta)$$

6) exponential check

$$P(n=0) = \lambda e^{-\lambda}$$

$$f(n, \lambda) = \lambda e^{-\lambda} \quad n > 0 \quad \lambda > 0$$

general flow:

$$f(n, \theta) = h(n) \exp[\eta(\theta) T(n) + A(\theta)]$$

$$f(n, \lambda) = \lambda e^{-\lambda}$$

$$= e^{\ln \lambda - \lambda}$$

$$h(n) = 1$$

$$\eta(\theta) = -\lambda$$

Normal check

$$P(n=n) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{(n-\mu)^2}{2\sigma^2} \right]$$

6) gamma check

$$f(n, \lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$$

$$= \frac{1}{\Gamma(\alpha)} \beta^\alpha \lambda^{\alpha-1} e^{-\beta\lambda}$$

expand using $(n-\mu)^2$

$$\frac{-n^2}{2\sigma^2} + \frac{\mu^2}{2\sigma^2} - \frac{2n\mu}{2\sigma^2}$$

$$\frac{1}{\Gamma(\alpha)} \exp(\ln \beta^\alpha + (\alpha-1) \ln \lambda - \beta\lambda)$$

$$\frac{1}{\Gamma(\alpha)} \exp(d \ln \beta + (\alpha-1) \ln \lambda - \beta\lambda)$$

$$= 1 = h(n)$$

$$T_2(\lambda) = \lambda$$

$$\ln(\lambda) = T_1(\lambda)$$

$$\eta_2(\theta) = \beta \quad A(\theta) = \alpha \ln \beta$$

probability is defined as a measure of happening of uncertainty.

Beta distribution

Sample space :-

$$T(\theta) = \frac{1}{\beta(\alpha, \beta)} e^{(\alpha-1) \ln \theta} e^{(\beta-1) \ln(1-\theta)}$$

The set of all possible outcomes of a random experiment

$$\frac{1}{\beta(\alpha, \beta)} \exp \left[(\alpha-1) \ln \theta + (\beta-1) \ln(1-\theta) \right]$$

Outcome space. Ω is a non empty set

The element of Ω are called elementary events

$$\exp(\alpha-1) \ln \theta + (\beta-1) \ln(1-\theta) - \ln \beta(\alpha, \beta)$$

Event space. Given an outcome space Ω an event space $\mathcal{E}$ over Ω

is a set of subset of Ω

$$h(\theta) = 1 \quad T_1(\theta) = \ln(\theta)$$

Ω is a set of subset of Ω

$$T_2(\theta) = \ln(1-\theta)$$

$$\eta_1(\theta) = \alpha - 1$$

Random Variable is a

$$\eta_2(\theta) = (\beta - 1)$$

numerical quantity whose value depends on outcome of a r. experiment

$$A(\theta) = \ln \beta$$

Graph generally represented as $G = (V, E)$

vertex

A vertex or node represents

an individual r.v

Edge.

represents a relationship or dependency b/w vertices

Two types of edges.→ Directed

has a specific direction,
 $x_i \rightarrow x_j$

→ undirected

No specific direction
 $x_i - x_j$

A graph $G(x, E)$ is directed if all edges $e \in E$ are directed

Further, a graph is said to be mixed.

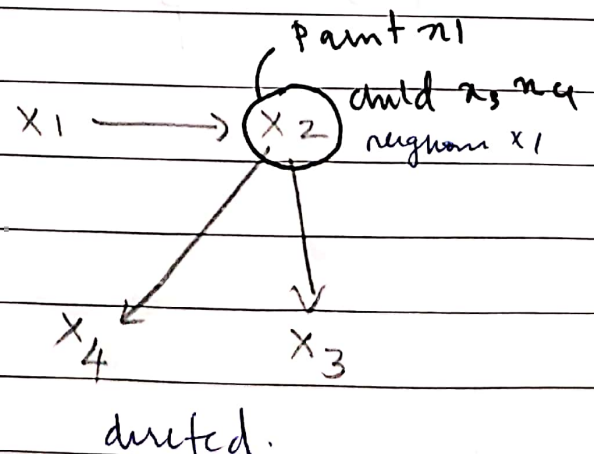
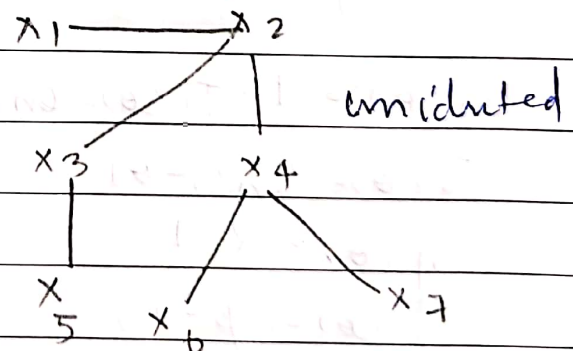
If the set of edge contains undirected and directed edge.

parent Node

If a Node has an outgoing arrow toward another node then it's called the parent node.

child Node.

A child node is the node that receives arrow from another node.



Neighbours.

Let G be mixed
or undirected graph.
and $u_i, u_j \in V$

$i \neq j$ if $(u_i, u_j) \in E$
then u_i and u_j said to be
a neighbour of u_j

Subgraph & Supergraph

Let $G = (V, E)$ and $G' = (V', E')$

be a graphs. A graph G'
called a subgraph of G if

$V \subseteq V'$ and $E \subseteq E'$

equivalently G' is called
a supergraph of G .

path

The sequence G is called
a path if every consecutive
pair of nodes is connected
by an edge.

Trail

A sequence of nodes is
called a trail

clique

In a graph $G(V, E)$
a set of nodes $C \subseteq V$
is a clique. If there
exist a graph edge between
all pairs of nodes.

Graph Theory in Probabilistic Graphical Model (PGMs)

Graphical Theory provides a mathematical framework to represent relationship among variables in a simple and visual form.

A graph is generally represented as:
 $G = (X, E)$

$X \rightarrow$ set of vertices

$E \rightarrow$ set of edges.

Vertex (Node)

A vertex or node represents an individual random variable in graphical model.

Edge

An edge represents a relationship or dependency between two vertices.

Two types of edges

i) undirected edge

$\rightarrow$ no direction, represented as
 $x_i - x_j$

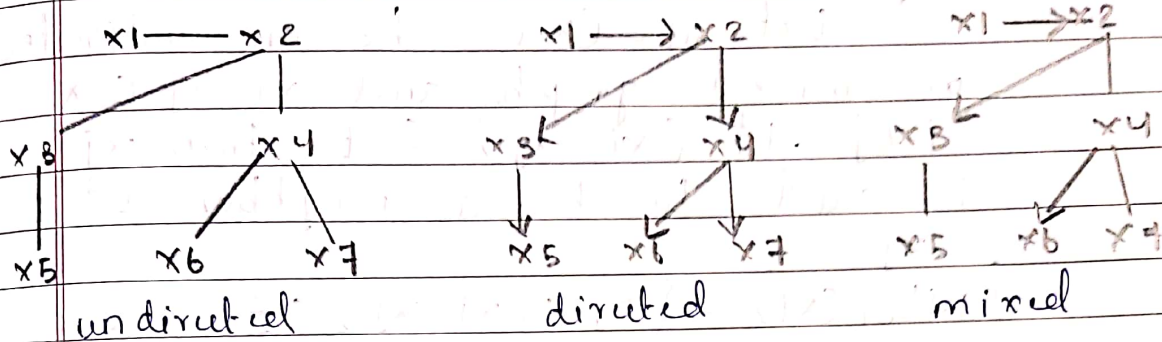
ii) Directed Edge

$\rightarrow$ has a specific direction, represented as:

$x_i \rightarrow x_j$

A graph $G = (X, E)$ is directed (undirected) if all edges $e \in E$ are directed (undirected). Further, a graph is said to be a mixed graph if the set of edge contains

undirected and directed edges



In graphs we can define relationships between variables.

Parent and child:

Let $G = (X, E)$ be a directed or mixed graph and $x_i, x_j \in X$, $i \neq j$. If $(x_i \rightarrow x_j) \in E$, then x_i is a parent of x_j and x_j is a child of x_i .

There exist $x_i \rightarrow x_j$

x_i is parent of x_j

x_j is called child of x_i

Parent Node

If a node has an outgoing arrow toward another node, then it is called the parent node.

eg: Rain $\rightarrow$ Wet Road

Then ; Rain is the parent
Wet Road is the child

child Node

A child node is the node that receives arrow from another node.

For undirected and mixed graphs we define the notion of neighborhood Neighbor:

Let $G = (X, E)$ be an undirected or mixed graph, and $x_i, x_j \in X$, $i \neq j$. If $(x_i - x_j) \in E$, then x_j is said to be a neighbor of x_i .

$$\{ Nbr(x_i) = \{ x_j : (x_i - x_j) \in E, x_j \in X \} \}$$

Path

The sequence α is called a path if every consecutive pair of nodes is connected by an edge.

i) Directed path

→ If atleast one edge in the path has a direction.

Trial

A sequence of nodes is called a trial if movement between consecutive nodes is possible through:

a forward directed edge
a backward directed edge
or an undirected edge.

$$(x_i \rightarrow x_{i+1}) \in E \vee (x_{i+1} \rightarrow x_i) \in E \vee (x_i - x_{i+1}) \in E$$

In a trial, edges are not repeated.

Clique And Maximal Clique.

In an undirected graph $G=(X,E)$ a set of nodes $C \subseteq X$ is a clique if there exists an edge between all pairs of nodes in the subset C ;
 $\forall c_i, c_j \in C, i \neq j : (c_i - c_j) \in E$

The clique C is a maximal clique if adding any node $x \in X \setminus C$ makes it no longer a clique.

Subgraph and Supergraph.

Let $G=(X,E)$
 and $G'=(X',E')$ be two graphs
 A graph G is called a subgraph of G' if

$$X \subseteq X' \text{ and } E \subseteq E'$$

Equivalently G' is called a supergraph of G .

Loop

A loop is an edge that connects a node to itself.
 $(x_i, x_i) \in E$
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